Yang-Mills Existence and Mass Gap via Adaptive Regularization

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Abstract

The Yang-Mills mass gap problem asks for a rigorous quantum field theory on $\mathbb{R}^4$ with compact simple gauge group G and a positive spectral gap $\Delta > 0$ above the vacuum. We analyze the structure of the problem via one-parameter regularization $E_\alpha[A] = E[A] + \alpha \int |F_{\mu\nu}|^4$. On a finite lattice, the regularized theory is rigorously defined and has a positive spectral gap for each $\alpha > 0$. Two open problems are identified explicitly: (i) the uniform-in-α lattice gap (equivalent to the lattice mass gap conjecture), and (ii) the continuum limit $a \to 0$. The classical non-abelian self-interaction provides a positive second variation at vacuum, supporting but not proving (i). The template closes conditionally on both.

1 Classical Setup

Let G be a compact simple Lie group with Lie algebra $\mathfrak{g}$. A connection A_μ on $\mathbb{R}^4$ has field strength:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

The Yang-Mills energy:

$$E[A] = \int_{\mathbb{R}^4} |F_{\mu\nu}|^2 \, d^4x \geq 0.$$

Vacuum: $F_{\mu\nu} = 0$ iff $A_\mu = g^{-1}\partial_\mu g$ (pure gauge). Configuration space: $\mathcal{A}/\mathcal{G}$.

2 Non-Abelian Self-Interaction

The field strength decomposes:

$$F^a_{\mu\nu} = (\partial_\mu A^a_\nu - \partial_\nu A^a_\mu) + f^{abc} A^b_\mu A^c_\nu$$

where f^{abc} are structure constants of $\mathfrak{g}$. For abelian G: $f^{abc} = 0$, no self-interaction. For non-abelian G: the commutator term is quartic in A.

Proposition 1 (Classical positive second variation)**.** *In the $k = 0$ topological sector, the second variation of E at vacuum is non-negative, with equality only on gauge zero modes.*

Proof. At $A = 0$, $\delta^2 E = \int |D_\mu \delta A_\nu - D_\nu \delta A_\mu|^2 \, d^4x \geq 0$, with $\delta^2 E = 0$ iff $\delta A_\mu = D_\mu \epsilon$ (gauge direction). On non-gauge directions, $\delta^2 E > 0$. □

Remark 2. Proposition 1 is classical, not quantum. The quantum spectral gap Δ is not simply the curvature of E at vacuum; it requires controlling the functional integral. The classical positivity is evidence for a positive quantum gap, but does not prove it.

3 Adaptive Regularization

Definition 3. For $\alpha > 0$: $E_\alpha[A] = E[A] + \alpha \int |F_{\mu\nu}|^4 \, d^4x$.

The $|F|^4$ term is superrenormalizable in 4d: it provides stronger UV suppression than the standard $|F|^2$ term.

4 Lattice Formulation

On a hypercubic lattice Λ with spacing $a > 0$ and $|\Lambda|$ sites, the gauge field is replaced by link variables $U_e \in G$ for each edge e. The regularized partition function:

$$Z_\alpha^\Lambda = \int_{G^{|\Lambda|}} \exp(-S_\alpha^\Lambda(U)) \prod_e dU_e$$

where dU_e is the Haar measure on G and S_α^Λ is the lattice action (Wilson plaquette plus α-weighted quartic term).

Lemma 4 (Lattice gap for $\alpha > 0$)**.** *For each $\alpha > 0$ and each finite lattice Λ, the transfer matrix T_α^Λ is a positive self-adjoint operator on a finite-dimensional Hilbert space $\mathcal{H}^\Lambda$. It has a spectral gap $\Delta_\alpha^\Lambda = \log(\lambda_0/\lambda_1) > 0$, where $\lambda_0 > \lambda_1$ are the two largest eigenvalues.*

Proof. $G^{|\Lambda|}$ is compact; the Haar integral defines a finite-dimensional operator. The quartic term $\alpha S_{\mathrm{qrt}}^\Lambda$ makes S_α^Λ strictly convex in a neighborhood of the identity, so T_α^Λ has a unique largest eigenvalue $\lambda_0 > \lambda_1$ by the Perron-Frobenius theorem applied to the positive integral kernel of T_α^Λ. □

Remark 5. Lemma 4 uses finite-dimensionality (compact group, finite lattice) in an essential way. The Brascamp-Lieb inequality [4] applies to log-concave measures in finite dimensions and could sharpen the gap estimate. In the continuum, Brascamp-Lieb is not directly applicable; no analogous bound in infinite-dimensional function space is known.

Conjecture 6 (Lattice mass gap conjecture, LMG)**.** *For the standard lattice Yang-Mills action ($\alpha = 0$) with non-abelian group G,*

$$\Delta_0^\Lambda \geq \Delta_0 > 0$$

uniformly over all lattice spacings $a > 0$ and volumes $|\Lambda|$.

Status of LMG. In the strong-coupling limit (large bare coupling g_0), the mass gap is established by cluster expansion methods (Osterwalder-Seiler, Seiler [5]). In the weak-coupling / continuum limit, LMG is open. Numerical lattice QCD strongly supports $\Delta_0 > 0$ for SU(2) and SU(3).

Lemma 7 (Tightness on lattice)**.** *For fixed Λ, the family $\{Z_\alpha^\Lambda\}_{\alpha \geq 0}$ is tight.*

Proof. For each $\alpha \geq 0$, Z_α^Λ is a probability measure on the compact space $G^{|\Lambda|}$. By Prokhorov's theorem, every such family is tight. □

Lemma 8 (Gap continuity in α, lattice)**.** *For fixed finite Λ, $\alpha \mapsto \Delta_\alpha^\Lambda$ is continuous. In particular, if $\Delta_0^\Lambda > 0$ then $\Delta_\alpha^\Lambda \geq \frac{1}{2}\Delta_0^\Lambda$ for all sufficiently small $\alpha \geq 0$.*

Proof. On a finite lattice, T_α^Λ depends analytically on α. Eigenvalues of a finite analytic family of self-adjoint operators are continuous. □

5 The Two Open Steps

The adaptive template produces two clean open problems:

1. **Uniform lattice gap (LMG, Conjecture 6).** Lemma 8 gives $\Delta_\alpha^\Lambda \geq c\Delta_0^\Lambda > 0$ for small α. Uniformity in the volume $|\Lambda|$ and lattice spacing a is Conjecture 6. Classical evidence: Proposition 1. Quantum evidence: strong-coupling expansions, numerics.

2. **Continuum limit.** Taking $a \to 0$ requires renormalization: the bare coupling $g_0(a)$ must run according to the β-function so that physical observables have finite limits. Controlling the spectral gap through the continuum limit is the hard analytic problem in constructive QFT [2].

6 Main Theorem (Conditional)

Theorem 9 (Yang-Mills existence and mass gap, conditional)**.** *Assume LMG (Conjecture 6) and that the continuum limit $a \to 0$ of the lattice Yang-Mills theory exists as a Wightman QFT. Then the quantum Yang-Mills theory on $\mathbb{R}^4$ with group G has a positive mass gap $\Delta \geq \Delta_0 > 0$.*

Proof. By Lemma 4, $\Delta_\alpha^\Lambda > 0$ for each $\alpha > 0$, Λ. Under LMG, $\Delta_0^\Lambda \geq \Delta_0 > 0$. By Lemma 8, $\Delta_\alpha^\Lambda \geq \frac{1}{2}\Delta_0 > 0$ for small α, uniformly in Λ. By tightness (Lemma 7) and the assumed continuum limit, the lattice measures converge to a continuum Wightman QFT Z_0. The spectral gap is preserved in the continuum limit by the assumed existence (which encodes renormalization). Therefore Z_0 has gap $\Delta \geq \frac{1}{2}\Delta_0 > 0$. □

Remark 10 (Template summary).

- *Regularize*: $E_\alpha[A]$ well-defined for $\alpha > 0$; gap exists on lattice. ✓
- *Bound*: Uniform gap $\Delta_\alpha^\Lambda \geq \Delta_0 > 0$. Conditional on LMG.

- *Compact*: Prokhorov tightness on compact lattice group. ✓
- *Regularity*: Gap preserved in continuum limit. Conditional on constructive QFT.

The two conditional steps are precisely the two layers of the Clay problem.

References

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